

Operation Illuminate — Architecture Explained

Admissions Analytics Platform Architecture, Purdue Expansion, and SES
Proxy Modeling

Thoughtmuseum — G Division

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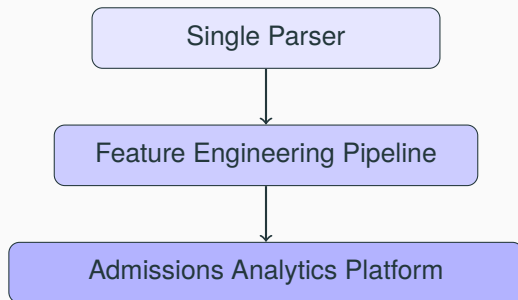
The project has evolved beyond:

single-school parser

into:

multi-school admissions analytics platform

The Purdue corpus should be treated as a second dataset instance operating on top of shared computational infrastructure.



Key Transition

CMU is no longer merely a completed parser project. It is the reference implementation for a reusable analytics framework.

CMU now functions as:

the canonical reference pipeline

The CMU project already contains:

- extraction infrastructure
- feature engineering systems
- NLP classification
- inferential analytics
- visualization pipelines
- reporting architecture

Core principle:

`shared/ = reusable machinery`

`cmu/, purdue/ = dataset instances`

This enables portability, reproducibility, schema compatibility, and future multi-school comparison.

CMU

- raw_pdfs/
- audit/
- extracted/
- features/
- analytics/
- graphics/
- reports/

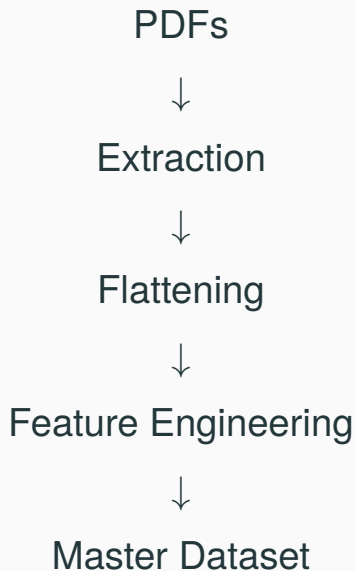
Purdue

- raw_pdfs/
- audit/
- extracted/
- features/
- analytics/
- graphics/
- reports/

Symmetry now becomes a design principle.

Core Architectural Layers

Layer	Purpose
Raw Data	PDFs and source application artifacts
Audit	PDF reconnaissance and parser validation
Extraction	Structured text extraction
Flattening	Activity normalization
Feature Engineering	Student-level modeling variables
Classification	NLP and semantic categorization
Analytics	Modeling and inferential statistics
Visualization	Graphics and interpretive plots
Reporting	Human-facing deliverables
Schema	Formal data contracts
Governance	Labels, lookups, review queues, documentation



Core responsibilities:

- PDF ingestion
- OCR detection
- section identification
- structured field extraction
- transcript parsing
- activity extraction

Primary engine:

`extract_structured_data.py`

Semi-structured activities require:

- normalization
- flattening
- category cleanup
- role decomposition
- activity standardization

Primary artifact:

`activities_flat.csv`

Feature engineering now includes:

- academic metrics
- research tiering
- award taxonomies
- activity leadership scores
- essay-theme indicators
- high-school features

Central analytical substrate:

`master_features_dataset.csv`

This dataset integrates academic variables, activities, awards, research, essays, and engineered applicant indicators.

The platform already contains:

AI-assisted semantic classification

Including:

- essay theme analysis
- research-title classification
- review queues
- human-in-the-loop adjudication

Current analytical modalities include:

- logistic regression
- ROC analysis
- odds-ratio analysis
- clustering
- feature differentiation
- inferential statistics

Emerging infrastructure includes:

- permutation tests
- cluster validation
- support thresholds
- robustness analysis
- future bootstrap workflows

The project is moving beyond descriptive analytics.

The visualization pipeline already produces:

- ROC curves
- coefficient significance plots
- odds-ratio forests
- permutation distributions
- essay-theme prevalence plots
- feature differentiation graphics

The platform now supports:

- milestone reports
- PDF export pipelines
- client-facing summaries
- graphics integration
- automated report generation

The existence of:

`shared/schemas/`

already represents major architectural maturity.

Schemas help prevent schema drift, undocumented features, modeling inconsistencies, and dataset incompatibility.

Purdue should begin with:

document anthropology

rather than immediate parser adaptation.

Goal

Discover which CMU assumptions fail under Purdue.

First target artifact:

```
purdue/audit/inventory/purdue_pdf_inventory.csv
```

Inventory variables:

- page count
- OCR likelihood
- text density
- section presence
- parser compatibility
- weirdness score

Folder	Purpose
raw_pdfs/	original Purdue application PDFs
audit/	reconnaissance, inspection, validation
extracted/	raw parser outputs
features/	engineered student-level datasets
analytics/	modeling and statistical outputs
graphics/	visualizations
reports/	human-facing reports
raw_labels/	labels, IDs, lookup tables
notes/	state files and parser assumptions

The client has proposed a new feature family:

sports, music, scouting, siblings, income proxies, and
activity-specific indicators

These features introduce a new modeling layer: applicant context and opportunity structure.

The project is beginning to move from:

direct observable feature modeling

into:

latent opportunity-structure inference

Potential SES Proxy Variables

Observable	Possible SES Signal
Zipcode	neighborhood income distribution
High school	opportunity structure
AP count	academic access
Private music training	family resources
Elite sports	travel and training resources
Fee waiver status	strong SES indicator
Parental occupation	major SES correlate
Part-time job	weak or ambiguous signal

Correct Statistical Framing

Avoid:

`infer true family income`

Instead:

`estimate posterior SES probabilities`

Example:

$\text{Pr}(\text{income} > \$250\text{k} \mid \text{observed features})$

Prior SES Distribution

+

Observed Applicant Signals

↓

Posterior SES Probabilities

Potential priors:

- census zipcode data
- school demographics
- regional income distributions
- institutional opportunity structures

Role of Bootstrapping

Bootstrapping does not infer income.

It estimates:

stability and uncertainty

Workflow:

1. Build SES proxy model
2. Resample applicants repeatedly
3. Recompute estimates
4. Measure variance and confidence intervals

The project should maintain:

- probabilistic framing
- uncertainty reporting
- exploratory interpretation
- careful documentation

And avoid:

- deterministic classification
- certainty claims
- direct income assertions
- overinterpretation

The project is evolving along two axes:

Engineering Maturity

- shared infrastructure
- schema discipline
- modular analytics
- reproducibility

Conceptual Sophistication

- latent-variable modeling
- opportunity structures
- Bayesian inference
- inferential analytics

Next steps:

1. Finalize CMU as reference implementation
2. Run Purdue PDF inventory
3. Validate CMU parser assumptions
4. Produce Purdue extracted datasets
5. Check schema compatibility
6. Build Purdue feature tables
7. Compare CMU and Purdue feature behavior

Operation Illuminate

The Purdue corpus is not merely additional data.

It is the first true scalability test of the AdmissionsAnalytics platform.

The architectural foundations are now strong.

Future effort should focus on:

controlled generalization rather than organizational rescue.